

The algebraic monodromy of Lagrangian fibrations

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Holomorphic symplectic manifolds

Definition

A *symplectic manifold* (M, ω) is a smooth manifold M equipped with a closed non-degenerate global 2-form $\omega \in H^0\left(M, \bigwedge^2 T^*M\right)$.

Contracting with ω induces an isomorphism $\iota_\omega : TM \xrightarrow{\sim} T^*M$.

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Note that in the latter two examples, despite X being a complex manifold, the Kähler form $\omega \sim dz_1 \wedge d\bar{z}_1 + \cdots + dz_n \wedge d\bar{z}_n$ is not holomorphic.

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- Any product of the spaces above is symplectic.

The decomposition theorem

We can generalize the K3 surface as follows:

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We say a compact holomorphic symplectic manifold (X, σ) is *irreducible* if it is simply connected and $\bigoplus_{p=0}^{\dim X} H^0(X, \Omega_X^{2p}) = \mathbb{C}[\sigma]$.

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Since complex tori are somewhat well understood, we will focus on understanding the irreducible holomorphic symplectic (IHS) X_i .

Definition

Let (X, σ) be a holomorphic symplectic manifold of dimension $2n$, and let $Y \subseteq X$ be a subvariety.

- 1 Y is *isotropic* if $\sigma|_Y = 0$.
- 2 Y is *Lagrangian* if Y is maximally isotropic, i.e. $\sigma|_Y = 0$ and $\dim Y = n$.

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For this reason, we refer to any such morphism $f : X \rightarrow B$ as a *Lagrangian fibration*. Understanding the geometry of Lagrangian fibrations will help us understand the geometry of IHS manifolds.

The structure of the fibration

Let $f^\circ : X^\circ \rightarrow B^\circ$ be the smooth fibers of $f : X \rightarrow B$. We need to understand

- the base B ,
- the general fiber X_b ,
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By Hodge theory, an Abelian variety A can be recovered from its cohomology:

$$A \simeq H^{1,0}(A)/(H^1(A, \mathbb{Z}))^\vee.$$

We will study the monodromy representation

$$\rho : \pi_1(B^\circ, b) \rightarrow \text{Aut}(H^1(X_b, \mathbb{Z})).$$

Algebraic monodromy groups

Definition

The *full algebraic monodromy group* M of ρ is the Zariski closure of $\rho(\pi_1(B^\circ, b)) \subseteq \text{Aut}(H^1(X_b, \mathbb{Q})) = \text{Sp}_{2n, \mathbb{Q}}$. Its identity component $M^\circ \subseteq M$ is the *(connected) algebraic monodromy group*.

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- 3 $M^\circ = \{1\}$, acting trivially on $H^1(X_b, \mathbb{Q})$.

Geometric consequences

In each of the cases above, we extract information regarding the geometry of X and $f : X \rightarrow B$.

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The case (3) is also somewhat well understood based on previous work:

Theorem (Kim–Laza–Martin '25, V '26)

When $M^\circ = \{1\}$, there exists an elliptic curve E for which the smooth fibers X_b are isogenous to E^n .

Proof of the main theorem.

Proof idea. We have the following isomorphism of bundles:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{T}_{X^\circ/B^\circ} & \longrightarrow & \mathcal{T}_{X^\circ} & \longrightarrow & f^*\mathcal{T}_{B^\circ} \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ 0 & \longrightarrow & f^*\Omega_{B^\circ}^1 & \longrightarrow & \Omega_{X^\circ}^1 & \longrightarrow & \Omega_{X^\circ/B^\circ}^1 \longrightarrow 0 \end{array}$$

The fibers of f are given by quotienting $f_*\Omega_{X^\circ/B^\circ,b}^1$ by a full rank sublattice. In particular, the structure of the fibers of f comes from $\mathcal{T}_B$.

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There exists a cover $\pi : B' \rightarrow B$ for which $\mathcal{T}_{B'} = \mathcal{F}_1 \oplus \cdots \oplus \mathcal{F}_k$ splits into transverse foliations $\mathcal{F}_i$, which are permuted by the covering group $\mathfrak{S}_k$.

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$$p : B' \rightarrow Z^k.$$

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Proof idea (continued). Now quotient $p : B' \rightarrow Z^k$ by the symmetric group to get $\bar{p} : B \rightarrow \operatorname{Sym}^k Z$. This induces a surjection on intersection cohomology:

$$\mathrm{IH}^*(\operatorname{Sym}^k Z, \mathbb{Q}) = \operatorname{Sym}^k (\mathrm{IH}^*(Z, \mathbb{Q})) \subseteq \mathrm{IH}^*(B, \mathbb{Q}).$$

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By Matsushita, $\text{IH}^*(B, \mathbb{Q}) = H^*(\mathbb{P}^n, \mathbb{Q})$. Therefore

$$\text{IH}^*(Z, \mathbb{Q}) = H^*(\mathbb{P}^1, \mathbb{Q}),$$

forcing $Z = \mathbb{P}^1$ and $k = 1$. □

Questions